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E-272 CLOSED CIRCUIT TELEVISION SYSTEM FOR SECURITY

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E-272 CLOSED CIRCUIT TELEVISION SYSTEM FOR SECURITY

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to site, installation, commissioning and testing of closed circuit television system for security in accordance with Part 4, Subsection 1.2 Summary of Work.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards and the following specific structures.

2.1 National

PUIL	:	Peraturan Umum Instalasi Listrik - Indonesia (General Regulation of Electrical Installation – Indonesia).
SPLN	:	Standard Perusahaan Umum Listrik Negara.
SII	:	Standard Industry Indonesia
EIA	:	Electronic Industry Association Standards.

2.2 International Electrotechnical Commission

IEC 60728	:	Cable distribution system primarily intended for sound and television signals operating between 30 MHz and 1 GHz.
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3. OPERATING DESIGN CONDITIONS

The equipment shall be designed for operation, in a tropical climate along the seacoast. Refer to Part 4, Subsection 1.1, 2.14 and 2.16 for meteorological, seismic and environmental data, at site.

The CCTV System shall be designed to operate in light conditions ranging from full sunlight to artificial illumination conditions as covered in Specification E-270. Camera placement shall consider ultimate substation from a security stand point in being able to observe movements within and near the power plant and switchyard for monitoring persons of a suspicious nature.

4. DETAILED REQUIREMENTS

4.1 General

The design of the CCTV system shall be such that steam turbine area, power plant perimeter Block 2, 3 and 4 area including switchyard extension, halls (exterior and interior) shall be readily visible by use of the CCTV system.

The CCTV system shall be completely installed system including the covers, cables, controls, monitors, lights.

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CCTV for Block 2 shall be a new system or extension of the existing Block 1 (whichever applicable) and will be operated from Control Room of Block 2. CCTV for Block 3 and 4 shall be separated and will be operated from Control Room of Block 3 and 4.

All equipment shall be such that local equivalents are available where possible for repairs, replacement and temporary usage. PAL system type signalling as found in Indonesia shall be used.

The CCTV system shall be high resolution color signalling system.

All cameras, video monitor and video amplifiers shall be powered from the same phase using 230 Volt AC, 50 Hz. Other equipment may use other phases in the same housing. All equipment using other voltages shall have internal or local controllers/transformers as required.

4.2 Cameras

Cameras shall be of the outdoor type for very low light surveillance standard considering the light incipient on the ground based on Specification E-270 requirements. Charge coupled (CCD type) pick-up designed for the specified ambient illumination shall be used.

Each camera used for general surveillance shall have a remotely controlled zoom lens of 10:1 ratio and an aperture control system to allow full day light with sun protection and full night time usage and remote focusing.

Each camera housing enclosure shall be designed to protect the camera from

environmental effects and damage. The features shall include but not be limited to the following:

- (1) Heavy duty housing with shrouded glass/plastic viewing port of sufficient strength.
- (2) Water resistant
- (3) Strip heater for anti-condensation
- (4) Viewing port windshield wiper with washer and 4 liter storage bottle
- (5) Remote control equipment for controlling all camera functions from master station
- (6) Auxiliary receptacle for small 230 V AC tools using local receptacle styles
- (7) Speaker for addressing from remote master station

Each general surveillance, camera housing enclosure shall be mounted on a pan-tilt unit for outdoor construction with the following range.

- (1) Operating range of 340° horizontal, 15° upward and 60° downward.
- (2) Pan and tilt speed of approximately 2.5 degrees per second.

Camera housings will be mounted on either buildings or separate towers at least 8 meters above grade level, located such that the viewing ranges overlap and protect the blind area of adjacent cameras.

Cameras shall be located such that a 1.5 meter high man occupies 5% of the monitor screen when standing, with zoom lens at twice minimum focal range.

Each exterior camera shall have spot lights for camera vision if sufficient light is not available during abnormal weather conditions and detailed surveillance. It is expected that the normal night lighting from switchyard general lighting is sufficient for general surveillance.

The Gate house camera shall have a fixed lens, outdoor housing same as Clause 4.2.3 and shall include external speaker-microphone for two way communication with approaching visitors.

4.3 Console Monitor and Controls

Two (2) duplicate consoles shall be provided for monitoring and controlling the security system for each separate system. One of these consoles shall be located in the power plant central control room and the other console shall be located in the security office room to be determined later. The Contractor shall be responsible for furnishing, installing, commissioning and testing of both consoles. The Contractor shall supply and install redundant communication link to both consoles.

Two (2) 499 mm LCD color video monitors shall be provided and mounted on each console desk. These monitors shall have high resolution picture with heavy duty equipment for continuous service.

Each monitor shall independently be able to receive any camera input and sequence through the camera inputs such that, while one monitor is sequencing the other may be held on any one camera, or both monitors can be held on different cameras.

For each monitor a remote control master for all cameras shall be provided using digital signalling techniques for all functions of the camera, including but not be limited to, the following:

- (1) Zoom
- (2) Near-far focusing
- (3) Pan (May be joystick control)
- (4) Tilt
- (5) Iris control, manual and automatic
- (6) Wiper and washer, on and off
- (7) Light on and off

The control masters shall operate in parallel, with first command override priority to any one camera (i.e. camera completes the commanded operation prior to responding to the other master command).

Supplementary amplifiers shall be supplied for video signals as required for visual picture enhancement where low light or long cable distances require such enhancement.

4.4 Cables

All video cabling shall be of coaxial design with moisture impervious sheath, using 75 ohm or equivalent cabling, with sufficiently low loss and with minimum 95% shield coverage.

The control signal cabling shall be provided as required for remote control of the camera and housing.

All cabling shall be suitably shielded for use in a 500 kV Substation, and sheathed against termite and vermin damage.

Power cabling shall be provided of sufficient size to meet voltage drop requirements.

4.5 Test Equipment

One set of complete testing and checkout equipment including extension cords, portable monitor, oscilloscope etc., shall be provided for testing all equipment such that testing equipment is at ground level when cameras are maintained.

4.6 Structures

All support structures and foundations shall conform to Civil/Structural Specifications.

4.7 Drawings

All drawings shall be in accordance with Part 4, Subsection 2.

Contractor shall provide with his proposal a site layout showing camera locations, area of visibility per the specifications, and blind spots due to camera movement limits.

5. PERFORMANCE TESTS AND GUARANTEES

Contractor shall provide all environmental testing to demonstrate that the components and the complete assembly shall meet the environmental characteristics of the Site. If previous tests are utilized, Contractor shall justify these tests in relation to Site conditions.

Contractor shall provide all routine tests for factory verification of equipment including a 30 day burn-in test to verify solid state component quality.

Contractor shall completely interconnect the system of cameras with housings and mounts, monitors and remote control system in the factory and functionally test the system operability prior to shipment.

Contractor shall functionally test and verify operability of the system after erection including normal night usage and completeness of perimeter scanning.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control for CCTV System for Security shall conform to the requirements included in Part 4, section 2, subsection 2.7.

The quality assurance/quality control procedures specified shall include, but not be limited to, the following:

- (1) All tests specified in Clause 5 of this specification.
- (2) All field installation activities associated with interconnecting the system.

The quality assurance/quality control documentation as specified shall include, but not be limited to, the following:

- (1) Shop test reports.
- (2) Environmental test reports.
- (3) Functional test reports for both manufacturing facility and at Site.

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